Random Intersections of Planes Generated by Uniform Points on the Sphere

Problem

Let U be uniform on the unit sphere $S^2 \subset \mathbb{R}^3$. Take three iid copies $x_1, x_2, x_3 \sim U$. Almost surely these determine a unique plane, which we write in oriented form as

$$P = \{x \in \mathbb{R}^3 : n \cdot x = h\}, \qquad n \in S^2, \quad h \in [-1, 1].$$

Now take three independent such planes

$$P_i = \{x : n_i \cdot x = h_i\}, \qquad i = 1, 2, 3,$$

constructed from disjoint triples of iid uniform points on S^2. Almost surely the planes meet at a unique point

$$X = P_1 \cap P_2 \cap P_3.$$

We want the distribution of

$$R = \|X\|.$$

Geometric Intuition

The problem combines two geometric phenomena. First, three random points on S^2 typically lie on a circle (the intersection of S^2 with a plane), and the density of planes is not uniform: planes close to the equator (where $|h|$ is small) are much more common than planes near the poles. This explains the factor $(1 - h^2)$ in the offset density $g(h)$.

Second, three planes in general position meet at a single point. The randomness in the three normals n_1, n_2, n_3 introduces randomness in both the location and distance of this intersection from the origin. The isotropy of the problem (invariance under rotations of $\mathbb{R}^3$) means that the distance R depends only on the norms, not on the directions, which is why the problem reduces to a radial density $f_R(r)$.

Summary of the Result

The key exact fact is the law of one random plane:

- its oriented normal n is uniform on S^2,
- n is independent of its signed offset h,
- the offset density is

$$g(h) = \frac{3}{4}(1 - h^2), \qquad -1 \le h \le 1.$$

From this, the intersection point density is

$$f_X(x) = \mathbb{E}\left[|\det(n_1, n_2, n_3)| \prod_{i=1}^{3} g(n_i \cdot x)\right].$$

Hence

$$f_X(x) = \left(\frac{3}{4}\right)^3 \mathbb{E}\left[|\det(n_1, n_2, n_3)| \prod_{i=1}^{3} (1 - (n_i \cdot x)^2)_+\right],$$

where $(a)_+ := \max(a, 0)$.

By isotropy, $f_X(x)$ depends only on $\|x\|$, and therefore

$$f_R(r) = 4\pi r^2 f_X(re_3), \qquad r \geq 0.$$

I do not presently have a closed elementary form for f_R, but the formulas below reduce it to an explicit low-dimensional integral.

1. Law of a Single Random Plane

Let σ denote surface area measure on S^2, and let $\mu = \sigma/(4\pi)$ be the uniform probability measure.

Take three iid uniform points $x_1, x_2, x_3 \in S^2$. Almost surely they determine a unique plane, which can be written as

$$P = \{x : n \cdot x = h\}.$$

For fixed $n \in S^2$ and $h \in (-1, 1)$, define

$$\rho = \sqrt{1 - h^2}.$$

Then the intersection

$$C(n, h) := S^2 \cap \{x : n \cdot x = h\}$$

is a circle of radius ρ.

Choose a rotation $R_n \in SO(3)$ with $R_n e_3 = n$, and parametrize the circle by

$$x_i = h\, n + \rho\, R_n v(\theta_i), \qquad v(\theta) = (\cos\theta, \sin\theta, 0), \qquad i = 1, 2, 3.$$

Thus

$$(n, h, \theta_1, \theta_2, \theta_3) \mapsto (x_1, x_2, x_3)$$

parametrizes almost every non-collinear triple on S^2.

By rotational invariance it suffices to compute the Jacobian at $n = e_3$. Put

$$v_i = v(\theta_i), \qquad w_i = (-\sin\theta_i, \cos\theta_i, 0).$$

Use local coordinates on S^2 near e_3 given by infinitesimal rotations about the e_1- and e_2-axes, with parameters a, b. Then the six tangent vectors in $(\mathbb{R}^3)^3$ are

$$\partial_{\theta_i} x = (0, \ldots, 0, \rho w_i, 0, \ldots, 0),$$

$$\partial_h x = (-\tfrac{h}{\rho} v_1 + e_3,\ -\tfrac{h}{\rho} v_2 + e_3,\ -\tfrac{h}{\rho} v_3 + e_3),$$

$$\partial_a x = (e_1 \times x_1,\ e_1 \times x_2,\ e_1 \times x_3),$$

$$\partial_b x = (e_2 \times x_1,\ e_2 \times x_2,\ e_2 \times x_3).$$

Let G be the 6×6 Gram matrix of these vectors. A direct Schur-complement computation gives

$$\det G = \rho^6 \det S,$$

where

$$S = \begin{pmatrix} 3/\rho^2 & (\sum \sin\theta_i)/\rho & -(\sum \cos\theta_i)/\rho \\ (\sum \sin\theta_i)/\rho & \sum \sin^2\theta_i & -\sum \sin\theta_i \cos\theta_i \\ -(\sum \cos\theta_i)/\rho & -\sum \sin\theta_i \cos\theta_i & \sum \cos^2\theta_i \end{pmatrix}.$$

This can be factorized as

$$S = \operatorname{diag}(\rho^{-1}, 1, 1)\, Z Z^\top \operatorname{diag}(\rho^{-1}, 1, 1),$$

with

$$Z = \begin{pmatrix} 1 & 1 & 1 \\ \sin\theta_1 & \sin\theta_2 & \sin\theta_3 \\ -\cos\theta_1 & -\cos\theta_2 & -\cos\theta_3 \end{pmatrix}.$$

Hence

$$\det G = \rho^4 (\det Z)^2.$$

Now

$$\frac{1}{2} |\det Z|$$

is the Euclidean area of the triangle in the plane with vertices

$$(\cos\theta_i, \sin\theta_i), \qquad i = 1, 2, 3.$$

Call that area $A(\theta_1, \theta_2, \theta_3)$. Therefore the Jacobian is

$$J = \sqrt{\det G} = 2\rho^2 A(\theta_1, \theta_2, \theta_3).$$

Thus the surface-area decomposition is

$$d\sigma(x_1)\, d\sigma(x_2)\, d\sigma(x_3) = 2(1 - h^2)\, A(\theta_1, \theta_2, \theta_3)\, d\sigma(n)\, dh\, d\theta_1\, d\theta_2\, d\theta_3.$$

After integrating out the circle angles, the dependence on h is exactly $1 - h^2$. Therefore the law of (n, h) has density

$$\mathbb{P}\big((n, h) \in d\sigma\, dh\big) = c(1 - h^2)\, d\sigma(n)\, dh$$

for some constant c. Since

$$\int_{S^2} d\sigma = 4\pi, \qquad \int_{-1}^{1} (1 - h^2)\, dh = \frac{4}{3},$$

normalization gives

$$c = \frac{3}{16\pi}.$$

So the exact one-plane law is

$$\boxed{\mathbb{P}\big((n, h) \in d\sigma\, dh\big) = \frac{3}{16\pi}(1 - h^2)\, d\sigma(n)\, dh}$$

or equivalently

$$\boxed{n \sim \mathrm{Unif}(S^2), \qquad n \perp h, \qquad g(h) = \frac{3}{4}(1 - h^2)\mathbf{1}_{[-1,1]}(h).}$$

2. Density of the Intersection Point

Let

$$N = \begin{pmatrix} n_1^\top \\ n_2^\top \\ n_3^\top \end{pmatrix}, \qquad H = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}.$$

Then almost surely

$$X = N^{-1}H.$$

Condition on N. Since h_1, h_2, h_3 are independent with density g,

$$f_{X|N}(x \mid N) = |\det N| \prod_{i=1}^{3} g(n_i \cdot x).$$

Taking expectation in N gives

$$\boxed{f_X(x) = \mathbb{E}\left[|\det(n_1, n_2, n_3)| \prod_{i=1}^{3} g(n_i \cdot x)\right].}$$

Substituting the explicit one-plane density,

$$\boxed{f_X(x) = \left(\frac{3}{4}\right)^3 \mathbb{E}\left[|\det(n_1, n_2, n_3)| \prod_{i=1}^{3} (1 - (n_i \cdot x)^2)_+\right].}$$

This is already an exact density formula.

3. Radial Reduction

By isotropy, $f_X(x)$ depends only on $r = \|x\|$, so we may set $x = re_3$.

Write

$$t_i := n_i \cdot e_3 \in [-1, 1], \qquad a_i := \sqrt{1 - t_i^2},$$

and parametrize

$$n_i = (a_i \cos\phi_i, a_i \sin\phi_i, t_i).$$

Since $d\sigma = dt\, d\phi$, one gets

$$f_X(re_3) = \frac{27}{2048\pi^2} \int_{[-1,1]^3} \Big(\prod_{i=1}^{3} (1 - r^2 t_i^2)_+\Big) K(t_1, t_2, t_3)\, dt_1 dt_2 dt_3,$$

where

$$K(t_1, t_2, t_3) = \int_0^{2\pi} \int_0^{2\pi} |D(u, v; t_1, t_2, t_3)|\, du\, dv,$$

with

$$D(u, v; t_1, t_2, t_3) = a_1 a_2 t_3 \sin u - a_1 a_3 t_2 \sin v + a_2 a_3 t_1 \sin(v - u).$$

Therefore

$$\boxed{f_R(r) = 4\pi r^2 f_X(re_3), \qquad r \geq 0.}$$

That is,

$$\boxed{f_R(r) = \frac{27\, r^2}{512\pi} \int_{[-1,1]^3} \Big(\prod_{i=1}^{3} (1 - r^2 t_i^2)_+\Big) K(t_1, t_2, t_3)\, dt_1 dt_2 dt_3.}$$

4. One More Angular Reduction

For fixed v, the quantity $D(u,v)$ is of the form

$$D(u,v) = A(v)\sin u + B(v)\cos u + C(v),$$

where

$$A(v) = a_1a_2t_3 - a_2a_3t_1\cos v,$$
$$B(v) = a_2a_3t_1\sin v,$$
$$C(v) = -a_1a_3t_2\sin v.$$

Let

$$R(v) = \sqrt{A(v)^2 + B(v)^2}.$$

Then

$$\frac{1}{2\pi}\int_0^{2\pi} |D(u,v)|\,du = \Psi(R(v), C(v)),$$

where

$$\Psi(R,C) = \begin{cases} |C|, & |C| \geq R, \\ \dfrac{2}{\pi}\left(\sqrt{R^2 - C^2} + |C|\arcsin\dfrac{|C|}{R}\right), & |C| < R. \end{cases}$$

Hence

$$K(t_1,t_2,t_3) = 2\pi\int_0^{2\pi} \Psi(R(v), C(v))\,dv.$$

So the radial density can be written as

$$\boxed{f_R(r) = \frac{27\,r^2}{256}\int_{[-1,1]^3}\Big(\prod_{i=1}^{3}(1 - r^2t_i^2)_+\Big)\left[\int_0^{2\pi}\Psi(R(v), C(v))\,dv\right]dt_1dt_2dt_3.}$$

5. Two Rigorous Asymptotic Results

The integral representation can be pushed further in two directions: the behavior near the origin, and the order of the tail at infinity.

5.1. Behavior near the Origin

At $x = 0$ the exact density simplifies to

$$f_X(0) = \left(\frac{3}{4}\right)^3 \mathbb{E}[|\det(n_1, n_2, n_3)|]\,.$$

So the first task is to compute the expectation of the random absolute determinant.

Fix n_1, n_2. Then

$$|\det(n_1, n_2, n_3)| = \|n_1 \times n_2\|\left|\frac{n_1 \times n_2}{\|n_1 \times n_2\|}\cdot n_3\right|.$$

Conditionally on n_1, n_2, the unit vector

$$\frac{n_1 \times n_2}{\|n_1 \times n_2\|}$$

is deterministic, while n_3 is uniform on S^2. Hence

$$\mathbb{E}\left[\left|\frac{n_1 \times n_2}{\|n_1 \times n_2\|} \cdot n_3\right| \middle| n_1, n_2\right] = \frac{1}{2},$$

because for a uniform point on S^2, the projection onto any fixed axis is uniform on $[-1, 1]$. Therefore

$$\mathbb{E}[|\det(n_1, n_2, n_3)|] = \frac{1}{2}\,\mathbb{E}[\|n_1 \times n_2\|]\,.$$

If Θ is the angle between n_1 and n_2, then $\cos\Theta$ is uniform on $[-1, 1]$, and

$$\|n_1 \times n_2\| = \sin\Theta = \sqrt{1 - \cos^2\Theta}.$$

Hence

$$\mathbb{E}[\|n_1 \times n_2\|] = \frac{1}{2}\int_{-1}^{1} \sqrt{1 - t^2}\,dt = \frac{\pi}{4}.$$

It follows that

$$\boxed{\mathbb{E}[|\det(n_1, n_2, n_3)|] = \frac{\pi}{8}.}$$

Consequently,

$$\boxed{f_X(0) = \left(\frac{3}{4}\right)^3 \frac{\pi}{8} = \frac{27\pi}{512}.}$$

Since the density formula is smooth in x near the origin, we obtain

$$f_X(x) = \frac{27\pi}{512} + O(\|x\|^2) \qquad (\|x\| \to 0),$$

and therefore

$$\boxed{f_R(r) = 4\pi r^2 f_X(re_3) = \frac{27\pi^2}{128} r^2 + O(r^4) \qquad (r \to 0).}$$

5.2. Tail Order at Infinity

Starting from

$$f_R(r) = \frac{27\,r^2}{512\pi} \int_{[-1,1]^3} \Big(\prod_{i=1}^{3} (1 - r^2 t_i^2)_+\Big) K(t_1, t_2, t_3)\,dt_1 dt_2 dt_3,$$

we note that for large r, the factor $(1 - r^2 t_i^2)_+$ restricts the integral to $|t_i| \le r^{-1}$. Make the change of variables

$$t_i = \frac{s_i}{r}.$$

Then

$$f_R(r) = \frac{27}{512\pi\,r} \int_{[-1,1]^3} \Big(\prod_{i=1}^{3} (1 - s_i^2)\Big) K\left(\frac{s_1}{r}, \frac{s_2}{r}, \frac{s_3}{r}\right)\,ds_1 ds_2 ds_3.$$

Now use the explicit formula for D:

$$D\left(u, v; \frac{s_1}{r}, \frac{s_2}{r}, \frac{s_3}{r}\right) = \frac{1}{r}\Big(s_3 \sin u - s_2 \sin v + s_1 \sin(v - u)\Big) + O(r^{-3}),$$

uniformly for $u, v \in [0, 2\pi]$ and $s_i \in [-1, 1]$, since

$$a_i = \sqrt{1 - \frac{s_i^2}{r^2}} = 1 + O(r^{-2}).$$

Therefore

$$K\left(\frac{s_1}{r}, \frac{s_2}{r}, \frac{s_3}{r}\right) = \frac{1}{r} L(s_1, s_2, s_3) + O(r^{-3}),$$

where

$$L(s_1, s_2, s_3) := \int_0^{2\pi} \int_0^{2\pi} |s_3 \sin u - s_2 \sin v + s_1 \sin(v - u)| \, du \, dv.$$

Substituting this into the expression for $f_R(r)$ gives

$$f_R(r) = \frac{A}{r^2} + O(r^{-4}), \qquad r \to \infty,$$

with

$$\boxed{A = \frac{27}{512\pi} \int_{[-1,1]^3} \Big(\prod_{i=1}^{3} (1 - s_i^2)\Big) L(s_1, s_2, s_3) \, ds_1 ds_2 ds_3.}$$

Integrating the density tail, we obtain

$$\boxed{\mathbb{P}(R > r) = \frac{A}{r} + O(r^{-3}), \qquad r \to \infty.}$$

In particular,

$$\boxed{\mathbb{E}[R] = \infty.}$$

So the heavy-tail exponent is rigorous, even though the explicit constant A remains to be evaluated.

5.3. Moments and Tail Dominance

Since $\mathbb{E}[R] = \infty$, we examine fractional moments. For $0 < p < 1$,

$$\mathbb{E}[R^p] = \int_0^\infty r^p f_R(r) \, dr = \int_0^1 r^p f_R(r) \, dr + \int_1^\infty r^p f_R(r) \, dr.$$

The first integral is finite because $f_R(r) = O(r^2)$ as $r \to 0$. For the tail integral, using $f_R(r) \sim A/r^2$,

$$\int_1^\infty r^p \cdot \frac{A}{r^2} \, dr = A \int_1^\infty r^{p-2} \, dr < \infty \quad \iff \quad p < 1.$$

Therefore $\mathbb{E}[R^p] < \infty$ for all $0 < p < 1$, with the tail contribution dominating:

$$\int_1^\infty r^p f_R(r) \, dr \sim \frac{A}{1 - p}, \quad p \nearrow 1.$$

The variance $\mathbb{E}[R^2] = \infty$, so the distribution is heavy-tailed without finite second moment.

6. Higher-Dimensional Generalization

The one-plane law and intersection problem generalize to $\mathbb{R}^d$ for any $d \geq 2$.

6.1. Planes in $\mathbb{R}^d$

In $\mathbb{R}^d$, a $(d-2)$-dimensional affine subspace (hyperplane) is given by

$$P = \{x \in \mathbb{R}^d : n \cdot x = h\}, \qquad n \in S^{d-1}, \quad h \in [-1, 1].$$

For three random points x_1, x_2, x_3 on S^{d-1}, the intersection

$$C(n, h) := S^{d-1} \cap \{x : n \cdot x = h\}$$

is a $(d-2)$-dimensional sphere of radius $\rho = \sqrt{1-h^2}$. The parametrization by angles and the Jacobian calculation generalize: the surface-area element factors as

$$d\sigma(x_1)\, d\sigma(x_2)\, d\sigma(x_3) \propto (1-h^2)^{(d-2)/2}\, dh\, d\sigma(n)\, d\Omega,$$

where $d\Omega$ is the measure on the space of angles. After integrating out angles,

$$\boxed{g_d(h) = c_d(1-h^2)^{(d-2)/2}, \quad -1 \le h \le 1,}$$

for a normalization constant c_d depending on d.

6.2. Intersection in $\mathbb{R}^d$

For d independent planes in $\mathbb{R}^d$, the intersection point is

$$X = N^{-1}H, \quad N = \begin{pmatrix} n_1^\top \\ \vdots \\ n_d^\top \end{pmatrix}, \quad H = \begin{pmatrix} h_1 \\ \vdots \\ h_d \end{pmatrix}.$$

By the same conditioning argument,

$$f_X(x) = \mathbb{E}\left[|\det(n_1, \ldots, n_d)| \prod_{i=1}^{d} g_d(n_i \cdot x) \right].$$

The isotropy and radial reduction proceed as before: the radial density is

$$f_R(r) = V_{d-1}(1) \cdot r^{d-1} f_X(re_1),$$

where $V_{d-1}(1) = 2\pi^{d/2}/\Gamma(d/2)$ is the surface area of S^{d-1}.

6.3. Tail Behavior in Higher Dimensions

By similar asymptotics, the tail of $f_R(r)$ scales as $r^{d-1} \cdot r^{-d} = r^{-1}$ for all $d \ge 2$. This suggests that the heavy-tail exponent is universal in d: the first moment of R diverges in all dimensions. However, the constant coefficient A_d and the local behavior near the origin do depend on d.

7. Open Problems and Future Work

The exact problem admits several natural extensions and open questions:

1. **Closed-form density:** Is there a closed-form expression for $f_R(r)$ in terms of classical special functions (hypergeometric, Bessel, etc.)? The nested integrals suggest the answer may be no, but a proof either way would be valuable.

2. **Evaluation of the constant A:** Compute the triple integral

$$A = \frac{27}{512\pi} \int_{[-1,1]^3} (1 - s_1^2)(1 - s_2^2)(1 - s_3^2) L(s_1, s_2, s_3)\, ds_1 ds_2 ds_3$$

   where $L(s_1, s_2, s_3)$ is defined in Section 5.2. Monte Carlo simulations suggest $A \approx 2/3$. Can this be proved exactly?

3. **Specific Probabilities:** Numerical evidence strongly suggests that the probability of the intersection point lying outside the unit sphere is

$$P(R > 1) = \frac{2}{\pi}.$$

   Is there a geometric reason or a simplification of the integral for $r = 1$ that yields this value? Similarly, for a single coordinate X_i, we observe $P(|X_i| > 1) = 1/3$.

4. **Finite-dimensional approximation of f_R:** Develop an asymptotic or perturbative expansion in terms of special functions valid over the entire range $r \geq 0$.

5. **Higher moments:** Compute or bound $\mathbb{E}[R^p]$ for $0 < p < 1$ explicitly in terms of the constant A.

6. **Random polytope connection:** Relate this problem to the classical random polytope literature (e.g., the convex hull of random points). What is the combinatorial structure of the arrangement of random planes?

7. **Non-uniform spacing:** How does the distribution of R change if the three points defining each plane are not uniformly distributed, but follow some other law on S^2?

8. Conclusion

The clean exact part of the problem is the one-plane law:

$$g(h) = \frac{3}{4}(1 - h^2)\mathbf{1}_{[-1,1]}(h).$$

From it one gets an exact integral formula for the density of the intersection point X, and hence for the distance $R = \|X\|$.

At present I do not see an elementary closed form for f_R. The remaining nontrivial piece is the angular average of the absolute determinant. However, the note now gives:

- the exact law of a single random plane,
- an exact integral formula for the density of the intersection point,

- the exact local asymptotic
$$f_R(r) = \frac{27\pi^2}{128} r^2 + O(r^4),$$
- the rigorous tail form
$$\mathbb{P}(R > r) = \frac{A}{r} + O(r^{-3}),$$
  with A given by an explicit integral,
- a characterization of fractional moments: $\mathbb{E}[R^p] < \infty$ for $p < 1$, with tail dominance,
- a natural generalization to higher dimensions $\mathbb{R}^d$, showing the offset density scales as $(1 - h^2)^{(d-2)/2}$,
- evidence that the heavy-tail exponent -1 (equivalently, $\mathbb{P}(R > r) \sim A/r$) is universal across dimensions.

The problem connects classical random geometry with the study of random plane arrangements and suggests further research directions both in computing the explicit constant A and in exploring variants and generalizations.